RehabSense: A Low-Cost Bilateral Wearable System with Mounting-Invariant Activity Recognition for ACL Rehabilitation Monitoring

Yash Trivedi, Darshil Zade, Tejal Sapkale, Yukta Myadam and Naina Kokate
Department of Computer Engineering, Vishwakarma Institute of Technology, Pune, India
trivediyash154@gmail.com

***Abstract*: Return-to-sport decisions after anterior cruciate ligament (ACL) reconstruction depend largely on between-limb symmetry, yet objective movement assessment remains confined to the laboratory, and low-cost inertial wearables degrade when the straps are re-attached at a different orientation. This paper presents RehabSense, a research platform built around a US$18 wearable in which one ESP32 samples an MPU6050 inertial measurement unit on each shank and two heel force sensors on a common 100 Hz clock. Monitoring is treated as estimation under unknown per-wearing sensor rotations. Each wearing is calibrated to recover the sensor bias, the neutral gravity direction and the dominant rotation axis, and motion is described by features that are provably invariant to sensor rotation while retaining posture through the calibrated neutral pose. The activity model was chosen by a selection rule fixed before any results were examined. In subject-independent evaluation on real bilateral public data, the deployed random forest reached a macro-F1 of 0.921 that remained unchanged under random re-mounting, whereas raw-axis features fell from 0.943 to 0.199; inference took 12.9 ms per window at the 95th percentile. Device-format replay reproduced 98.4% of training-pipeline predictions, the asymmetry and movement-quality metrics responded monotonically to simulated deficits, and a cross-dataset test (macro-F1 of 0.15 to 0.24) sizes the domain gap for the planned on-device pilot.**

***Index Terms*: anterior cruciate ligament, wearable sensors, inertial measurement unit, human activity recognition, orientation invariance, limb symmetry, random forest, Internet of Things**

I. Introduction

Return-to-sport decisions after anterior cruciate ligament (ACL) reconstruction rely heavily on between-limb symmetry: in the Delaware-Oslo cohort, patients who returned before meeting objective criteria, including a limb symmetry index of at least 90%, had a markedly higher reinjury rate [1]. Symmetry is, however, measured objectively only in laboratories [2], so most of a recovery that lasts nine months or longer goes unobserved. Remotely supervised programs achieve early outcomes comparable to brochure-based home rehabilitation [3], but they seldom quantify how the patient moves.

Inertial measurement units (IMUs) are the most widely used portable modality for ACL-related assessment, yet most reported systems remain at technology readiness levels 3 to 6 [2], [4]. Three issues limit their use at home. First, estimators and activity models that assume a fixed sensor orientation degrade when the strap is re-attached at each session [5], [6]. Second, record-wise data splits overestimate accuracy [7], and public-dataset results are often assumed to carry over to new hardware despite well-documented domain shift [8], [9]. Third, composite recovery scores often combine components with weights that are neither justified nor reported.

RehabSense was designed with these issues in mind. Its main contributions are:

1) a low-cost bilateral node in which one ESP32 samples both shank IMUs and the force sensors in the same 100 Hz tick, with a stream-integrity layer that reports data loss, jitter, clock drift and sensor faults instead of filling them in;
2) a per-wearing calibration and a feature representation that is provably invariant to sensor rotation and still retains posture through the calibrated neutral pose;
3) a subject-independent benchmark on two public datasets with device-specific robustness tests and a model-selection rule fixed in advance;
4) asymmetry and movement quality index (MQI) metrics that report their components and confidence and never impute missing inputs.

II. Related Work and Research Gap

A. Wearable Sensing after ACL Reconstruction

Scoping reviews of 49 out-of-lab ACL studies [2] and 32 wearable-device studies [4] found few systems validated outside the laboratory and none integrated into routine care. Phan et al. [10] monitored 26 patients at home with five IMUs and found that inter-limb knee-extension asymmetry decreased from 7.7° to 2.6° between three and nine months after surgery, a change that laboratory testing captured only in part. Seel et al. [11] avoided assumptions about sensor orientation by identifying the joint axis from the motion. Low-cost ESP32 systems estimate knee flexion from thigh and shank MPU6050 sensors [12], add force sensors and a cloud back end for symmetry indices [13], or fuse IMU data at the edge with an RMSE of 3.1° to 6.4° against video [14].

B. Orientation-Robust and Domain-Robust Activity Recognition

Yurtman and Barshan [5] removed the effect of absolute sensor orientation by transforming the raw signals, Gil-Martín et al. [15] projected the signals onto a consistent reference

TABLE I
REPRESENTATIVE APPROACHES AND THE GAP ADDRESSED BY REHABSENSE

Approach	Cost	Out-of-lab	Sync. bilat.	Mount. robust	Activity model†	Transp. metrics
Laboratory systems [2]	High	✗	✓	n/a	✗	✓
Free-living IMUs [10]	Med.	✓	✓	∼	✗	∼
Low-cost knee devices [12], [13], [14]	Low	✓	∼	∼	✗	∼
Invariant HAR [5], [6], [15]	n/a	n/a	✗	✓	✓	✗
RehabSense (ours)	**Low**	✓	✓	✓	✓	✓

✓ addressed; ∼ partially or depending on the system; ✗ not addressed; n/a: not applicable.
†Learned activity model evaluated in a subject-independent manner.

frame, and Barua et al. [16] used heuristic invariant features. Baek et al. [6], motivated by repeated re-attachment at home, built invariance to independent per-location rotations into an SO(3)-equivariant network. Invariant representations, however, discard segment posture, the main cue for distinguishing sitting from standing, and generalization across datasets remains difficult even with self-supervised pretraining [8], [9].

C. Research Gap

As Table I shows, low-cost knee wearables have concentrated on single-joint angles, whereas orientation-robust activity recognition has been developed without rehabilitation metrics. To the best of our knowledge, no low-cost system combines (a) synchronous bilateral capture, (b) mounting-invariant activity recognition that preserves posture, (c) subject-independent, pre-registered evaluation with a measured domain gap, and (d) movement metrics with explicit components and missing inputs.

III. SYSTEM DESIGN

A. Wearable Node

An ESP32 DevKit V1 reads two GY-521 MPU6050 IMUs, strapped to the lateral left and right shanks, over one I^{2}C bus at 400 kHz (Fig. 2); the AD0 pin sets their addresses to 0x68 (left) and 0x69 (right). Both IMUs are configured for ± 4 g and ± 500°/s with a 44 Hz low-pass filter, verified by register readback at boot. Two FSR402 heel sensors form 10 kΩ dividers on ADC1 pins, since ADC2 is unavailable while Wi-Fi is active, and their output is treated as an uncalibrated load proxy rather than a force in newtons. An 18650 cell feeds a TP4056 charger and an MT3608 5 V boost converter; the parts cost about INR 1,550 (US$18).

A FreeRTOS task on core 1 reads both IMUs and the force channels in the same tick, assigns one microsecond timestamp and queues the sample in a lock-free ring buffer, and the main loop sends ten samples per WebSocket frame, optionally over TLS. A failed read is sent as `null` rather than a stale or other-side value, and status frames report buffer overflows and I^{2}C errors so that device losses can be separated from network losses.

Algorithm 1 Online processing of one device stream

Input: stream $(t_k, \mathbf{x}_k^{\mathrm{L}}, \mathbf{x}_k^{\mathrm{R}}, f_k)$; calibration $(\mathbf{b}_\omega, \kappa, \mathbf{g}_0, \mathbf{u})$ per IMU; model M with W and τ

for each received packet **do**
  check order, duplicates and gaps; mask invalid values as NaN
  correct bias and scale; update the tilt θ_t using (6)
  if a stride s has elapsed **then**
    $X \leftarrow$ last W seconds, anti-aliased and resampled to 25 Hz
    if more than 10% of X is missing **then**
      emit INSUFFICIENT_DATA
    else
      $\boldsymbol{\phi} \leftarrow$ 324 invariant features of X, from (3) and (4)
      $\mathbf{p} \leftarrow M(\boldsymbol{\phi})$
      emit $\arg\max \mathbf{p}$ if $\max \mathbf{p} \geq \tau$, else LOW_CONFIDENCE
      emit A using (8), or SINGLE_SIDE
    end if
  end if
  on a completed repetition: ROM proxy, SPARC and MQI (9)
  store results with provenance; publish them to the dashboard
end for

B. Protocol and Software Platform

A *hello* frame declares each IMU (side, placement and range), each force channel and a per-device key, and the provenance of the session is fixed at this handshake: a registered device presenting its own key is LIVE, the simulator is SIMULATED, and any other source is UNVERIFIED. Only LIVE data may be used for training. The service (Fig. 1) runs FastAPI and a Next.js dashboard on Vercel with PostgreSQL, whose LISTEN/NOTIFY mechanism relays live messages between instances. Raw samples (about 19 MB per hour) are kept for 30 days unless the patient consents to training use.

IV. METHODOLOGY

A. Problem Formulation

Each sample k carries a device timestamp t_k, a measurement $\mathbf{x}_k^s = [\mathbf{a}_k^s; \boldsymbol{\omega}_k^s] \in \mathbb{R}^6$ (g, °/s) for each side $s \in \{\mathrm{L}, \mathrm{R}\}$, and force proxies f_k. Every wearing applies an unknown rotation $\mathbf{R}^s \in SO(3)$ to each sensor, and each MPU6050 adds its own bias and scale error. For every window, the system produces (i) an activity $y \in \mathcal{Y}$, $|\mathcal{Y}| = 9$, or an abstention; (ii) a range-of-motion (ROM) proxy per repetition; (iii) an asymmetry score $A \in [0, 1]$; and (iv) an MQI in $[0, 100]$, each with a status and a confidence. The outputs must be invariant to $\mathbf{R}^s$, a missing sensor or side is never imputed or mirrored (only short gaps inside a window are interpolated), and the device path must apply exactly the training-time processing. Algorithm 1 summarizes the online procedure.

B. Stream Integrity

The server measures the sampling rate instead of trusting the declared value: $\hat{f} = 1/\operatorname{median}(\Delta t)$, while the effective rate $N_{\mathrm{rx}}/(t_{\mathrm{last}} - t_{\mathrm{first}})$ also reflects the losses revealed by sequence-number gaps. An IMU is considered frozen after 25 identical readings and saturated at 98% of full scale, and invalid samples are masked rather than repaired. For clock drift, the packet

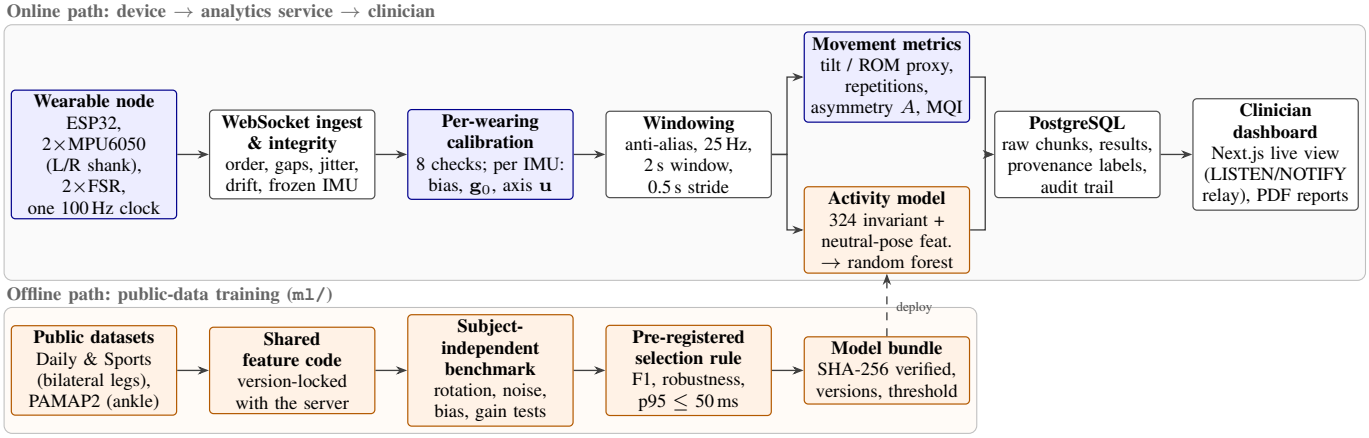


Fig. 1. Architecture of RehabSense. In the online path, the wearable node streams synchronized bilateral samples, and the service checks stream integrity, calibrates each wearing, segments the signals into windows and computes the activity estimate and the movement metrics; results reach the clinician dashboard through a PostgreSQL relay. In the offline path, candidate models are benchmarked on public data with the same feature code that runs on the server, selected by a pre-registered rule and deployed as hash-verified bundles.

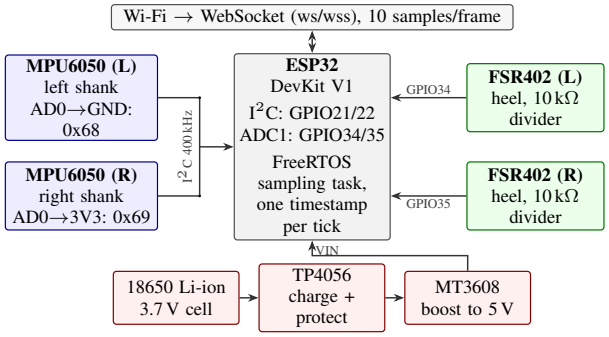


Fig. 2. Wearable node. A single ESP32 samples both shank IMUs (shared I^2C bus with distinct addresses) and the two heel force sensors in the same tick.

with the smallest delay $t^{\mathrm{arr}} - t^{\mathrm{dev}}$ is kept in each 5 s bucket b, and

$$t_b^{\mathrm{arr}} = \beta\, t_b^{\mathrm{dev}} + c, \qquad \delta = (\beta - 1) \cdot 10^6 \text{ ppm} \tag{1}$$

is fitted by least squares over at least 30 s. Fitting this lower envelope keeps network queuing from appearing as drift, and $|\delta| > 2000$ ppm marks the stream as not real-time.

C. Per-Wearing Calibration

Stillness is detected rather than assumed: the server waits for a 3 s span $\mathcal{S}$ in which the largest per-axis standard deviation of every healthy IMU stays below 0.03 g and 2.5°/s (timeout 20 s). From $\mathcal{S}$ it estimates the gyroscope bias, neutral gravity direction and accelerometer scale of each sensor:

$$\mathbf{b}_\omega = \overline{\boldsymbol{\omega}}_{\mathcal{S}}, \qquad \mathbf{g}_0 = \overline{\mathbf{a}}_{\mathcal{S}} / \|\overline{\mathbf{a}}_{\mathcal{S}}\|, \qquad \kappa = 1/\|\overline{\mathbf{a}}_{\mathcal{S}}\|. \tag{2}$$

During 5 s of slow repetitions $\mathcal{M}$, the rotation axis is the first right-singular vector of the bias-corrected angular velocities, $\boldsymbol{\Omega}_{\mathcal{M}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, $\mathbf{u} = \mathbf{v}_1$, provided their RMS exceeds 6°/s; its sign follows the direction of the first movement, which keeps it consistent across sessions. Table II lists the eight checks. The

TABLE II
PER-WEARING CALIBRATION CHECKS AND THRESHOLDS FOR MPU6050-CLASS SENSORS

Check	PASS	WARN (otherwise FAIL)
1 IMU connectivity	$\geq 95\%$ valid, not frozen	n/a
2 Sampling rate	$\lvert\hat{f} - f_{\mathrm{decl}}\rvert / f_{\mathrm{decl}} \leq 5\%$	$\leq 15\%$
3 Gravity magnitude	$\lVert\overline{\mathbf{a}}\rVert \in [0.9, 1.1]$ g	$[0.8, 1.2]$ g
4 Accel. baseline	max. SD ≤ 0.02 g	≤ 0.05 g
5 Gyro. baseline	noise $\leq 1°$/s, $\lvert\mathbf{b}_\omega\rvert \leq 20°$/s	noise $\leq 3°$/s
6 Force channel	load response‡	no load response
7 Neutral pose	$\mathbf{g}_0$ stored	n/a
8 Baseline movement	RMS $\geq 6°$/s, $\mathbf{u}$ found	axis learned later

‡SD during movement $> \max(3\sigma_{\mathrm{rest}}, 0.01)$; FAIL if the rest value is at least 0.98 of full scale.

calibration quality q_{cal} is the mean check score (PASS $= 1$, WARN $= 0.5$, FAIL $= 0$), and $\mathcal{S}$ and $\mathcal{M}$ are stored so that a calibration can be recomputed from raw data.

D. Mounting-Invariant Representation

After bias and scale correction, one function shared by training and deployment anti-aliases each window with a moving average and resamples it linearly to 25 Hz. Windows last $W = 2$ s with a stride of $s = 0.5$ s (response delay $\approx W/2 + s = 1.5$ s). With the window's gravity direction $\hat{\mathbf{g}} = \overline{\mathbf{a}}/\|\overline{\mathbf{a}}\|$, each IMU is mapped to six channels:

$$\begin{aligned} &\|\mathbf{a}_t\|, \quad \mathbf{a}_t \cdot \hat{\mathbf{g}}, \quad \|\mathbf{a}_t - (\mathbf{a}_t \cdot \hat{\mathbf{g}})\hat{\mathbf{g}}\|, \\ &\|\boldsymbol{\omega}_t\|, \quad \boldsymbol{\omega}_t \cdot \hat{\mathbf{g}}, \quad \|\boldsymbol{\omega}_t - (\boldsymbol{\omega}_t \cdot \hat{\mathbf{g}})\hat{\mathbf{g}}\|. \end{aligned} \tag{3}$$

These channels carry no posture information, which is needed to separate sitting from standing, so posture is measured relative to the calibrated direction $\mathbf{g}_0$ in the same sensor frame:

$$\theta_t^n = \arccos(\mathbf{a}_t \cdot \mathbf{g}_0 / \|\mathbf{a}_t\|)\,. \tag{4}$$

Proposition 1: For any rotation $\mathbf{R} \in SO(3)$ of a sensor that is fixed within a window, the channels in (3) and the neutral angle in (4) are unchanged.

Proof: The rotation maps $\mathbf{a}_t$, $\boldsymbol{\omega}_t$, $\hat{\mathbf{g}}$ (by linearity of the mean) and $\mathbf{g}_0$ (measured by the same sensor) to $\mathbf{R}\mathbf{a}_t$, $\mathbf{R}\boldsymbol{\omega}_t$, $\mathbf{R}\hat{\mathbf{g}}$ and $\mathbf{R}\mathbf{g}_0$. Since $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$, all dot products and norms are preserved, including $\|\mathbf{a} - (\mathbf{a}\cdot\hat{\mathbf{g}})\hat{\mathbf{g}}\|$ and $\mathbf{a}\cdot\mathbf{g}_0$; the same holds for $\boldsymbol{\omega}$. ∎

Each channel is summarized by 24 statistics (moments, percentiles, RMS, skewness, kurtosis, zero-crossing rate, dominant frequency, spectral entropy, four band powers, and autocorrelation peak and lag) plus the jerk RMS. Posture adds eight features (statistics of θ^n, and acceleration along and rotation about $\mathbf{g}_0$), and cross-side correlation, lag and variance-ratio features add 18, giving 153 features per IMU and 324 for the bilateral model, all invariant by Proposition 1.

E. Activity Recognition and Pre-Registered Selection

The classifier distinguishes the nine activities shared by the public datasets (lying, sitting, standing, walking, stairs up and down, running, cycling and other exercise). No public dataset contains squats, and since the prescribed exercise is known from the session, the classifier supplies free-living context. The feature-based candidates were a random forest (RF) with 200 trees, $\sqrt{d}$ features per split, minimum leaf size 2 and class-balanced bootstrap sampling [17], [18]; histogram gradient boosting (HGB); and the RF trained with device-like augmentation ($2\times$ noise, $\pm2°$/s bias, $\pm3\%$ gain). The deep candidates encode each IMU's nine-channel sequence (the channels of (3) plus three neutral-pose channels) with a shared three-layer 1D CNN, optionally followed by an LSTM [19], and fuse the embeddings symmetrically as $\mathbf{z} = [\mathbf{h}_\mathrm{L}, \mathbf{h}_\mathrm{R}, |\mathbf{h}_\mathrm{L} - \mathbf{h}_\mathrm{R}|, \mathbf{h}_\mathrm{L} \odot \mathbf{h}_\mathrm{R}]$, so the prediction does not depend on which leg wears which sensor; legs are also swapped with probability 0.5 in training. The networks (58.5k and 91.8k parameters) use class-weighted cross-entropy, AdamW (2×10^{-3}, cosine schedule, at most 15 epochs) and early stopping on a held-out training subject; one variant pretrains its encoder on PAMAP2.

The selection rule was fixed before any result was examined. For each model family m and window $W \in \{1, 2, 3\}$ s,

$$W_m^* = \min\{W : \mathrm{F1}_m(W) \geq \max_{W'} \mathrm{F1}_m(W') - 0.01\},$$
$$m^* = \arg\max_m \mathrm{F1}_m(W_m^*) \;\; \text{s.t.} \;\; \Delta_m^{\mathrm{rot}} \leq 0.03,\; L_m^{95} \leq 50\,\mathrm{ms}, \tag{5}$$

where Δ^{rot} is the macro-F1 drop under random rotation and L^{95} the 95th-percentile latency. Ties within 0.01 favor the smaller model, and a deep model is adopted only if it exceeds m^* by at least 0.02 under the same constraints. At run time the model abstains (LOW_CONFIDENCE) when $\max_c p(c \mid \boldsymbol{\phi}) < \tau = 0.3$.

F. Segment Tilt and Range-of-Motion Proxy

With a single IMU per leg a knee angle cannot be computed, so we estimate the shank tilt about the calibrated axis $\mathbf{u}$, an approach related to the joint-axis identification of [11]. The accelerometer tilt θ_t^a is the signed angle between $\mathbf{g}_0$ and $\mathbf{a}_t$ projected onto the plane normal to $\mathbf{u}$, fused with the gyroscope by a complementary filter [20]:

$$\theta_t = \alpha\,(\theta_{t-1} + (\boldsymbol{\omega}_t\cdot\mathbf{u})\,\Delta t) + (1-\alpha)\,\theta_t^a, \qquad \alpha = 0.98. \tag{6}$$

When $\|\mathbf{g}_0 - (\mathbf{g}_0\mathbf{u})\mathbf{u}\| < 0.3$, gravity carries no tilt information, and the estimate relies on the gyroscope alone with a slow decay toward neutral and is flagged.

G. Repetitions and Smoothness

The tilt is sign-normalized so that repetitions are positive excursions from the neutral pose. A peak is accepted if its prominence is at least $\max(A_{\min}, 0.35\,(P_{95} - P_5))$ and it lies at least $T_{\min}$ after the previous one ($A_{\min} = 10°$, $T_{\min} = 0.8$ s for repetitions; $8°$ and 0.6 s for gait cycles). A repetition spans the troughs around its peak, trimmed where the trace leaves and regains a 5% band around the trough level, and is kept only if the signal returns by at least half its rise. Its range is the ROM proxy, and its phases are labeled at 10%, 30% and 90% of the amplitude. Smoothness is the spectral arc length [21] of the angular-speed profile,

$$\mathrm{SPARC} = -\int_0^{\omega_c} \sqrt{\Big(\frac{1}{\omega_c}\Big)^2 + \Big(\frac{d\hat{V}(\omega)}{d\omega}\Big)^2}\, d\omega, \tag{7}$$

where $\hat{V}$ is the magnitude spectrum normalized to its peak and $\omega_c \leq 10$ Hz is the highest frequency at which $\hat{V} \geq 0.05$.

H. Bilateral Asymmetry

The two sides are compared through the normalized difference

$$d(L, R) = |L - R| / \max(|L|, |R|) \in [0, 1], \tag{8}$$

applied to the dynamic acceleration RMS, the RMS and 95th percentile of the angular velocity, and the ROM proxy, together with waveform dissimilarity (one minus the peak cross-correlation) and timing. For a per-leg period P and cross-correlation lag ℓ, the folded lag $\varphi = |((\ell + P/2) \bmod P) - P/2|$ gives a timing component of $\varphi/(P/2)$ for synchronous exercises such as squats and $|\varphi - P/2|/(P/2)$ for alternating ones such as walking. A is the mean of the available components, excluding those below the noise floor; if either side is available for less than 90% of the window, the system reports SINGLE_SIDE instead of a mirrored score. The confidence of A is $\min(1, T/4\,\mathrm{s})\cdot\min_s \alpha_s \cdot \sqrt{n_A/6}$ for window length T, side availability α_s and n_A available components.

I. Movement Quality Index

The MQI combines six components $c_i \in [0, 1]$: symmetry $(1 - A)$; the ROM proxy relative to the patient's baseline, capped at 1; temporal consistency (one minus the coefficient of variation of repetition durations); smoothness (median SPARC mapped linearly from $[-7, -1.5]$ to $[0, 1]$); force-peak consistency; and the mean pairwise correlation of time-normalized repetition profiles:

$$\mathrm{MQI} = 100\,\textstyle\sum_{i\in\mathcal{A}} w_i c_i \big/ \sum_{i\in\mathcal{A}} w_i \tag{9}$$

over the available components $\mathcal{A}$, with at least three repetitions. As no labeled data exist yet for fitting, the weights are equal, which we state as an assumption. The MQI confidence is $\sqrt{|\mathcal{A}|/6}\cdot\min(1, n_{\mathrm{rep}}/8)\cdot q_{\mathrm{stream}}\cdot(0.4 + 0.6\,q_{\mathrm{cal}})$, with values

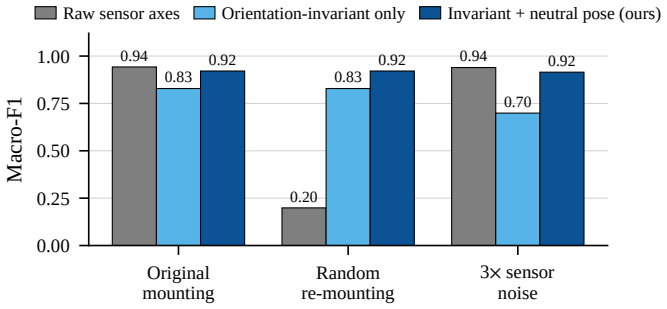


Fig. 3. Feature-set ablation for the bilateral RF with 2 s windows (subject-independent macro-F1) under the original mounting, random re-mounting and three-fold sensor noise.

below 0.5 reported as LOW_CONFIDENCE. The definition is frozen by a SHA-256 hash, and MQI values from different component sets are never compared.

V. Experimental Setup

Datasets. Daily and Sports Activities [22] provides 8 subjects with Xsens units recorded simultaneously on both legs and trains the bilateral model (60,424 windows). PAMAP2 [23] adds 9 subjects with an ankle IMU to the single-IMU model and to encoder pretraining. Single-IMU recordings were never mirrored into artificial left and right pairs, and one standing segment per subject emulated calibration and was excluded from evaluation.

Protocol. We used four-fold GroupKFold cross-validation by subject, with windows cut within recordings, normalization fitted on training subjects only and out-of-fold predictions pooled; macro-F1 was the primary metric [7]. Robustness was tested by perturbing raw test windows with a uniformly random rotation (applied to the signals and $\mathbf{g}_0$), Gaussian noise of 0.03 g and 1.5°/s (three times nominal), a per-axis gyroscope bias from $U(-3, 3)$°/s, an accelerometer gain from $U(0.95, 1.05)$ and a 5% time stretch. Transfer was measured between the Daily and Sports legs and the PAMAP2 ankle, and latency for features and prediction on one window, with one laptop CPU thread over 200 runs.

Verification. Public windows were replayed as quantized, jittered 100 Hz device streams, and a simulator attached each IMU at a random orientation to generate five exercises with known ROM, left-side deficits between 0 and 0.6, and injected faults.

VI. Results and Discussion

A. Mounting Invariance

Fig. 3 confirms Proposition 1 empirically. Raw-axis features score highest with the original mounting (0.943) but fall to 0.199 once the test sensors are rotated, as happens when a strap is re-attached. Invariant channels alone are unaffected by rotation, yet they lose posture information (0.829) and are sensitive to noise (0.699). Adding the neutral-pose features restores 0.921, with no loss under rotation and a small loss under three-fold noise (0.915).

TABLE III
Subject-independent benchmark (Daily and Sports, 9 classes)

Model (features)	W (s)	F1	Folds $\mu \pm \sigma$	Rot.	3×noise	p95 (ms)
RF (raw axes, control)	2	0.943	0.940 ± 0.019	**0.199**	0.939	5.4
RF (invariant only)	2	0.829	0.824 ± 0.049	0.829	0.699	5.0
RF (inv. + neutral)⋆	**2**	**0.921**	$\mathbf{0.907 \pm 0.061}$	**0.921**	**0.915**	**12.9**
RF (inv. + neutral)	3	0.929	0.916 ± 0.063	0.929	0.919	5.3
RF (+ device aug.)	2	0.921	0.908 ± 0.061	0.921	0.919	6.2
HGB (inv. + neutral)	3	0.945	0.935 ± 0.069	0.945	0.852	156.4
1D CNN	2	0.871	0.863 ± 0.045	0.871	0.868	0.4
CNN-LSTM	2	0.896	0.892 ± 0.030	0.896	0.893	0.9
CNN, pretrained	2	0.898	0.885 ± 0.027	0.898	0.900	0.3
RF, single IMU	2	0.895	0.894 ± 0.017	0.895	0.859	5.0

⋆Deployed model. F1: pooled out-of-fold macro-F1; Folds: per-fold macro-F1 over 4 folds; Rot.: after random sensor rotation. Pretrained: encoder pretrained on PAMAP2. Single IMU: each Daily and Sports leg plus the PAMAP2 ankle. Transfer from Daily and Sports to PAMAP2: 0.241; reverse: 0.153.

B. Model Selection

Table III summarizes the benchmark. HGB achieves the highest macro-F1 (0.945 with 3 s windows) but needs 156 ms per window, about three times the budget. The RF with 2 s windows (0.921) lies within 0.01 of its family's best and responds faster, so rule (5) selects it; the augmented RF ties but loses the tie-break on size (15.4 against 9.6 MB). Per-fold scores vary by about ± 0.06, mainly because one held-out group of subjects is harder (0.831 for the RF), so the 0.024 gap between HGB and the RF is smaller than the variation between folds and latency decides the selection. The deep models score 0.023 to 0.051 lower, although PAMAP2 pretraining improves the CNN by 0.027 with less than half the RF's fold variability, making it the preferred candidate for force fusion. The deployed model needs 4.7 ms per window on the server, but its 220 MB footprint precludes on-device inference.

C. Per-Class Behavior and Selective Prediction

Over the 60,424 out-of-fold windows, cycling, running, standing and lying exceed a recall of 0.99 and other exercise reaches 0.98. About a quarter of sitting windows are labeled lying, the posture pair that one segment per leg separates least reliably, and some walking and stair windows fall into other exercise, which includes stepper and cross-trainer movements. With $\tau = 0.3$ the model labels 99.99% of windows at 91.8% accuracy, and $\tau = 0.7$ gives 97.3% on 80.2% of windows; the threshold must be re-derived on device data.

D. Domain Gap and Consistency of the Device Path

Cross-dataset transfer reduces macro-F1 to 0.241 and 0.153, quantifying the cost of a change in sensor and placement, so public-data scores are evidence for pretraining rather than estimates of device accuracy. The device path is consistent with the training path: device-format replay reproduces the training-path prediction for 98.4% of 800 windows, and the full processor agrees with the labels for 95.6% of 1,152 windows. Since these replays use subjects seen in training, they verify integrity, not generalization.

E. Movement Metrics in Simulation

Against simulated ground truth, the median tilt error stays below 2.5° under random strap orientations, repetition counts lie within one to three of the simulated cycles (incomplete cycles are excluded), and all four injected faults (packet loss, IMU dropout, a frozen IMU and impossible values) are detected. The ROM proxy is within $\pm 1.5\%$ for the four repetitive exercises but underestimates walking by 9.4%, probably because heel strikes violate the quasi-static assumption behind θ_t^a. As the injected deficit grows from 0 to 0.6, the squat asymmetry score rises monotonically from 0.01 to 0.26 (0.08 to 0.40 for walking) and the MQI falls from 96.0 to 91.0, 88.5 and 80.8, with a repeatability of ± 0.08 over ten seeds. Removing the right IMU changes the MQI from 96.0 to 95.2 solely because the symmetry component disappears, which is why such values are never compared.

VII. Limitations and Future Work

RehabSense is a research prototype and not a medical device. The activity model has been validated on public data only; it has not yet been evaluated on labeled recordings from the device, and the transfer results suggest that accuracy will drop until the model is adapted. The ROM proxy is a segment tilt rather than a knee angle, the force channels are an uncalibrated load proxy, and the MQI weights and SPARC bounds are assumptions characterized only in simulation. The next stage is an ethics-approved pilot with consenting participants recorded on two days with re-strapping, therapist labels, zero-shot and fine-tuned leave-one-subject-out evaluation, goniometry and an intraclass-correlation analysis of MQI reliability. By lowering the cost of objective home monitoring, the work also supports UN Sustainable Development Goals 3 and 9.

VIII. Conclusion

This paper presented RehabSense, a US$18 bilateral wearable with an analytics pipeline designed for unsupervised use at home. Measuring posture relative to a calibrated neutral pose keeps the features provably invariant to sensor rotation without sacrificing posture recognition: the deployed model maintains a macro-F1 of 0.921 under random re-mounting, against 0.199 for raw-axis features. The platform is ready for its on-device pilot study.

Acknowledgment

The authors thank the Department of Computer Engineering, Vishwakarma Institute of Technology, Pune, for its support, and the creators of the Daily and Sports Activities and PAMAP2 datasets for making their data public.

References

[1] H. Grindem, L. Snyder-Mackler, H. Moksnes, L. Engebretsen, and M. A. Risberg, "Simple decision rules can reduce reinjury risk by 84% after ACL reconstruction: The Delaware-Oslo ACL cohort study," *Br. J. Sports Med.*, vol. 50, no. 13, pp. 804-808, 2016.

[2] T. Tan, A. A. Gatti, B. Fan, K. G. Shea, S. L. Sherman, S. D. Uhlrich, J. L. Hicks, S. L. Delp, P. B. Shull, and A. S. Chaudhari, "A scoping review of portable sensing for out-of-lab anterior cruciate ligament injury prevention and rehabilitation," *npj Digit. Med.*, vol. 6, p. 46, 2023.

[3] J.-Y. Lim *et al.*, "Effectiveness of in-home, augmented reality-based telerehabilitation after anterior cruciate ligament reconstruction: A randomized controlled trial," *Orthop. J. Sports Med.*, vol. 12, 2024.

[4] O. Ptaszyk, T. Boutefnouchet, G. Cummins, J. M. Kim, and Z. Ding, "Wearable devices for the quantitative assessment of knee joint function after anterior cruciate ligament injury or reconstruction: A scoping review," *Sensors*, vol. 25, no. 18, p. 5837, 2025.

[5] A. Yurtman and B. Barshan, "Activity recognition invariant to sensor orientation with wearable motion sensors," *Sensors*, vol. 17, no. 8, p. 1838, 2017.

[6] S. Baek, Y. Chai, Y. Lee, S. Choi, and S. Suh, "Rotation-invariant multi-IMU activity recognition under independent per-location orientation shifts," in *Proc. ACM Int. Symp. Wearable Comput. (ISWC)*, Shanghai, China, 2026.

[7] S. Saeb, L. Lonini, A. Jayaraman, D. C. Mohr, and K. P. Kording, "The need to approximate the use-case in clinical machine learning," *GigaScience*, vol. 6, no. 5, p. gix019, 2017.

[8] Z. Hong, Z. Li, S. Zhong, W. Lyu *et al.*, "CrossHAR: Generalizing cross-dataset human activity recognition via hierarchical self-supervised pretraining," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 8, pp. 1-26, 2024.

[9] O. Napoli, D. Duarte, P. Alves, D. H. P. Soto, H. E. de Oliveira, A. Rocha, L. Boccato, and E. Borin, "A benchmark for domain adaptation and generalization in smartphone-based human activity recognition," *Sci. Data*, vol. 11, 2024.

[10] V. Phan *et al.*, "The gait lab effect: Symmetry restoration strategy after anterior cruciate ligament reconstruction is different in natural environments than the gait laboratory," *J. Biomech.*, 2026.

[11] T. Seel, J. Raisch, and T. Schauer, "IMU-based joint angle measurement for gait analysis," *Sensors*, vol. 14, no. 4, pp. 6891-6909, 2014.

[12] J. Korbel *et al.*, "GyroStep: A portable IMU-based system for knee flexion measurement and real-time analysis," in *Proc. 30th Int. Conf. Methods Models Autom. Robot. (MMAR)*, 2026.

[13] J. Joshita K *et al.*, "A real-time IoT-enabled wearable system for knee rehabilitation using IMU-based angle estimation and force sensor analytics," in *Proc. 2nd Int. Conf. Innov. Mech., Robot., Comput. Biomed. Eng. (ICIMRBE)*, 2026.

[14] A. Achmamad *et al.*, "IoT-based system using IMU sensor fusion for knee telerehabilitation monitoring," *IEEE Sensors J.*, 2025.

[15] M. Gil-Martín, J. López-Iniesta, F. Fernández-Martínez, and R. San-Segundo, "Reducing the impact of sensor orientation variability in human activity recognition using a consistent reference system," *Sensors*, vol. 23, no. 13, p. 5845, 2023.

[16] A. Barua, X. Jiang, and D. Fuller, "The effectiveness of simple heuristic features in sensor orientation and placement problems in human activity recognition using a single smartphone accelerometer," *BioMed. Eng. OnLine*, vol. 23, p. 21, 2024.

[17] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5-32, 2001.

[18] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825-2830, 2011.

[19] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.

[20] R. Mahony, T. Hamel, and J.-M. Pflimlin, "Nonlinear complementary filters on the special orthogonal group," *IEEE Trans. Autom. Control*, vol. 53, no. 5, pp. 1203-1218, 2008.

[21] S. Balasubramanian, A. Melendez-Calderon, A. Roby-Brami, and E. Burdet, "On the analysis of movement smoothness," *J. NeuroEng. Rehabil.*, vol. 12, p. 112, 2015.

[22] K. Altun, B. Barshan, and O. Tunçel, "Comparative study on classifying human activities with miniature inertial and magnetic sensors," *Pattern Recognit.*, vol. 43, no. 10, pp. 3605-3620, 2010.

[23] A. Reiss and D. Stricker, "Introducing a new benchmarked dataset for activity monitoring," in *Proc. 16th Int. Symp. Wearable Comput. (ISWC)*, Newcastle, U.K., 2012, pp. 108-109.